

```

1  # Note 15: Hierarchial Clustering
2  # =====
3
4  # Import Libraries
5  from sklearn.cluster import AgglomerativeClustering
6  import numpy as np
7  import matplotlib.pyplot as plt
8  import scipy.cluster.hierarchy as hc
9
10 # =====
11 # 01. HC Agglomerative Clustering Model
12 # =====
13
14 # Define Data
15 X = np.array([1, 2, 3, 5, 6, 8, 9]).reshape(-1, 1)
16
17 # Create a Model
18 model = AgglomerativeClustering(n_clusters=3, linkage="ward")
19
20 # Fit & Predict From Model
21 labels = model.fit_predict(X)
22
23 # Print Predictions
24 print(labels)
25
26 # =====
27 # 02. Dendrogram
28 # =====
29
30 X = np.array([1, 2, 3, 5, 6, 8, 9])
31 y = np.array([2, 3, 1, 7, 8, 7, 9])
32
33 h = list(range(len(X)))
34
35 for i, v in enumerate(h):
36     plt.annotate(v, xy=(X[i], y[i]))
37
38 data = list(zip(X, y))
39
40 z = hc.linkage(data, method='ward')
41
42 plt.figure()
43 hc.dendrogram(z)
44 plt.xlabel('Data Points')
45 plt.ylabel('Distance')
46 plt.title('Hierarchical Clustering Dendrogram')
47 plt.show()

```